Lecture 1

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Math 308 E, F

Book: Jeffrey Holt, "Linear Algebra w/ Applications" (2nd ed., w/ Web Assign)

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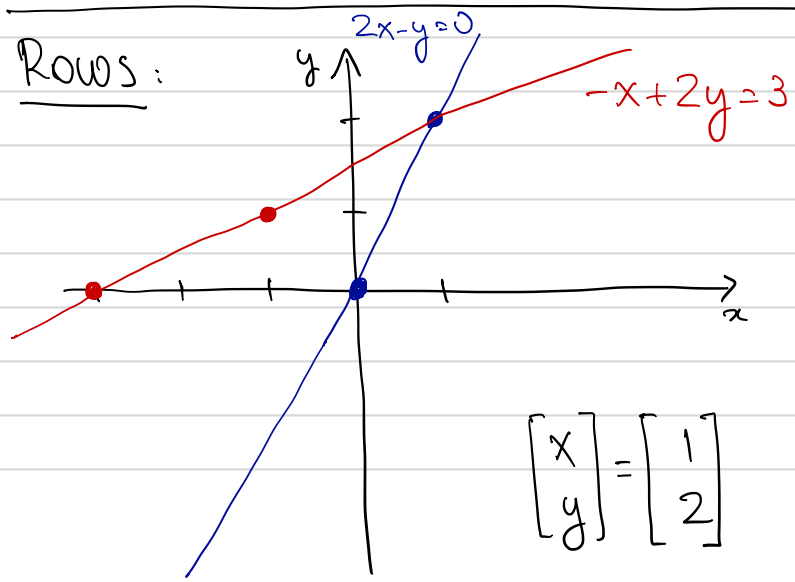
Today:
- n eqn's, n var's
- interpretation by rows
- " " columns (*)
- Gaussian Elimination

$$2x - y = 0$$
$$-x + 2y = 3$$

$$\underbrace{\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 0 \\ 3 \end{bmatrix}}_{b}$$

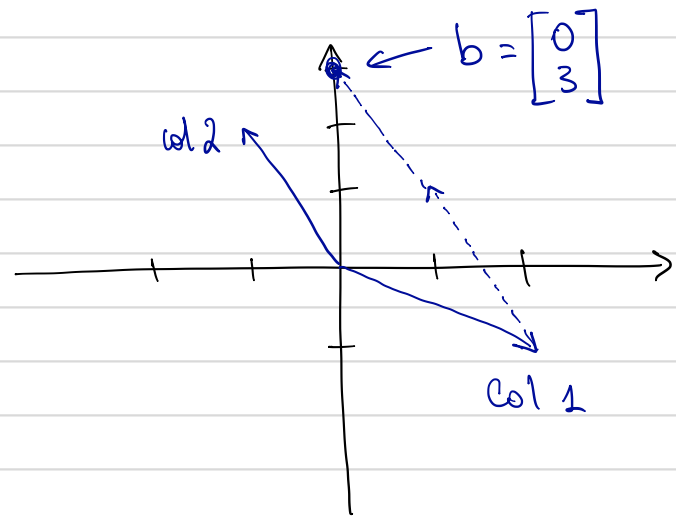
Rows:



$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Columns:

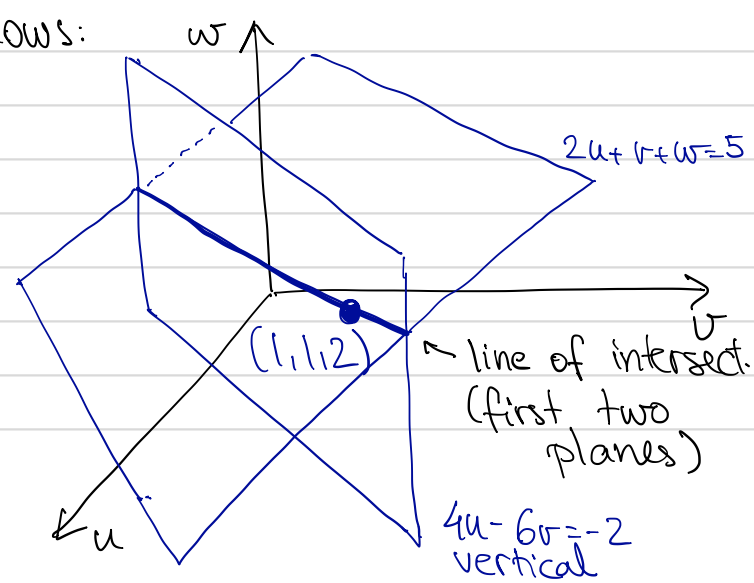
$$x \underbrace{\begin{bmatrix} 2 \\ -1 \end{bmatrix}}_{\text{col 1}} + y \underbrace{\begin{bmatrix} -1 \\ 2 \end{bmatrix}}_{\text{col 2}} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$



$$2u + v + w = 5$$
$$4u - 6v = -2$$
$$-2u + 7v + 2w = 9$$

$$\underbrace{\begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} u \\ v \\ w \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}}_{b}$$

Rows:



Columns:

$$u\begin{bmatrix}2\\4\\-2\end{bmatrix}+v\begin{bmatrix}1\\-6\\7\end{bmatrix}+w\begin{bmatrix}1\\0\\2\end{bmatrix}=\begin{bmatrix}5\\-2\\9\end{bmatrix}$$

$$1\cdot\begin{bmatrix}2\\4\\-2\end{bmatrix}+1\cdot\begin{bmatrix}1\\-6\\7\end{bmatrix}+2\cdot\begin{bmatrix}1\\0\\2\end{bmatrix}=\begin{bmatrix}5\\-2\\9\end{bmatrix} \checkmark$$

Gaussian Elimination

$$\begin{aligned}
2u + v + w &= 5\\
4u - 6v &= -2\\
-2u + 7v + 2w &= 9
\end{aligned}$$

$$\Big\downarrow \begin{array}{l} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 + R_1 \end{array}$$

$$\begin{aligned}
2u + v + w &= 5\\
-8v - 2w &= -12\\
8v + 3w &= 14
\end{aligned}$$

$$\Big\downarrow R_3 \to R_3 + R_2$$

$$\begin{aligned}
2u + v + w &= 5\\
-8v - 2w &= -12\\
1\cdot w &= 2
\end{aligned}$$

Back substitution:

$$\boxed{w=2,\quad v=1,\quad u=1}$$